2026.
11.5

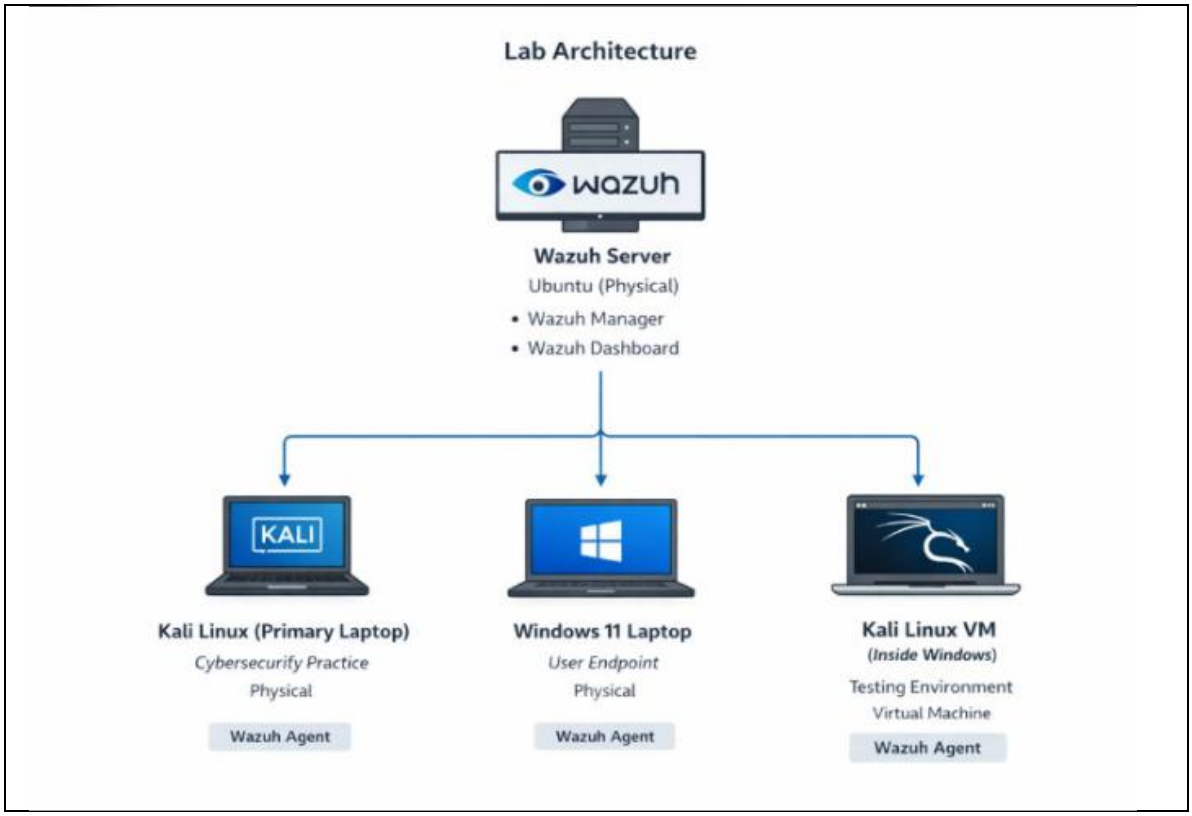
LINUX SERVER SECURITY MONITORING SYSTEM

AAKIF M.M.A.

Today's practical project

- Beginner → Intermediate

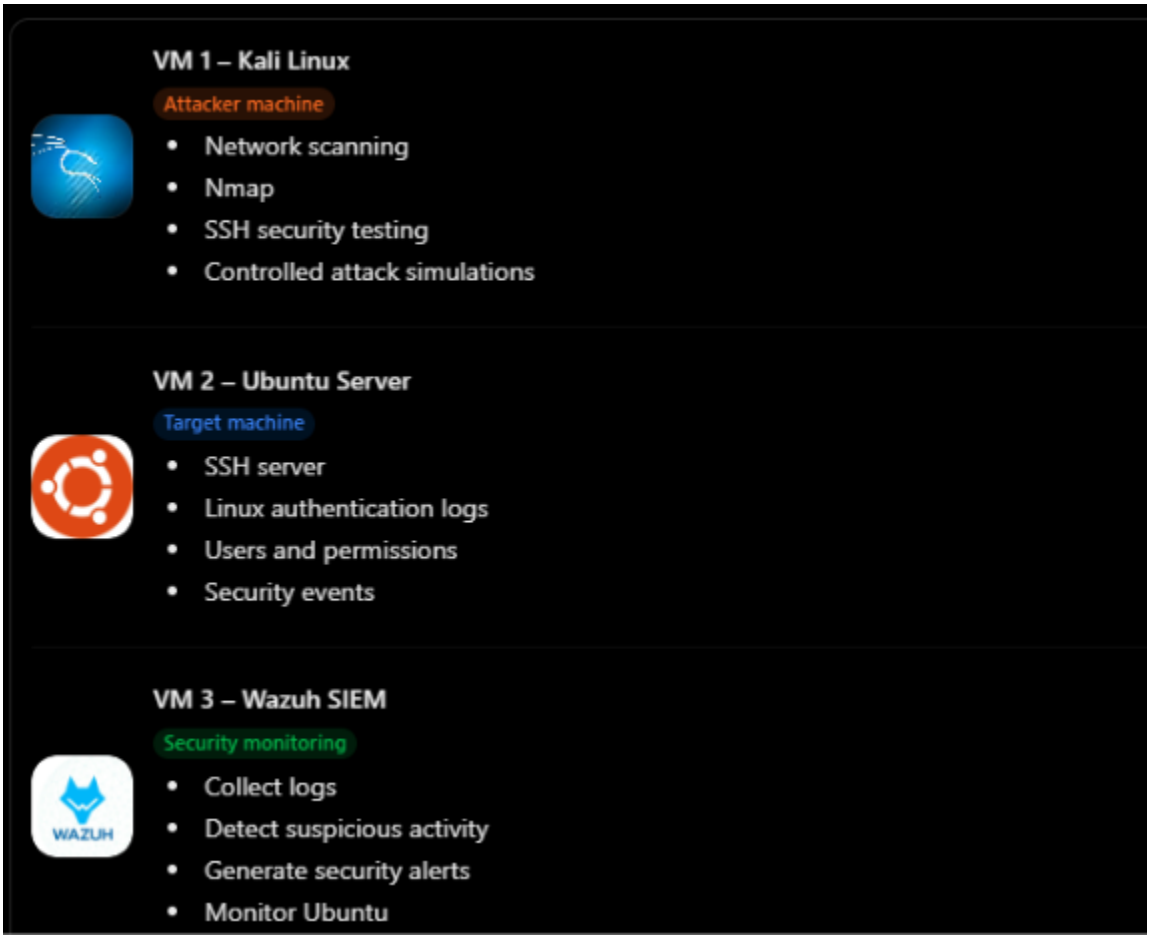
1. Project architecture



Your lab will have these machines:

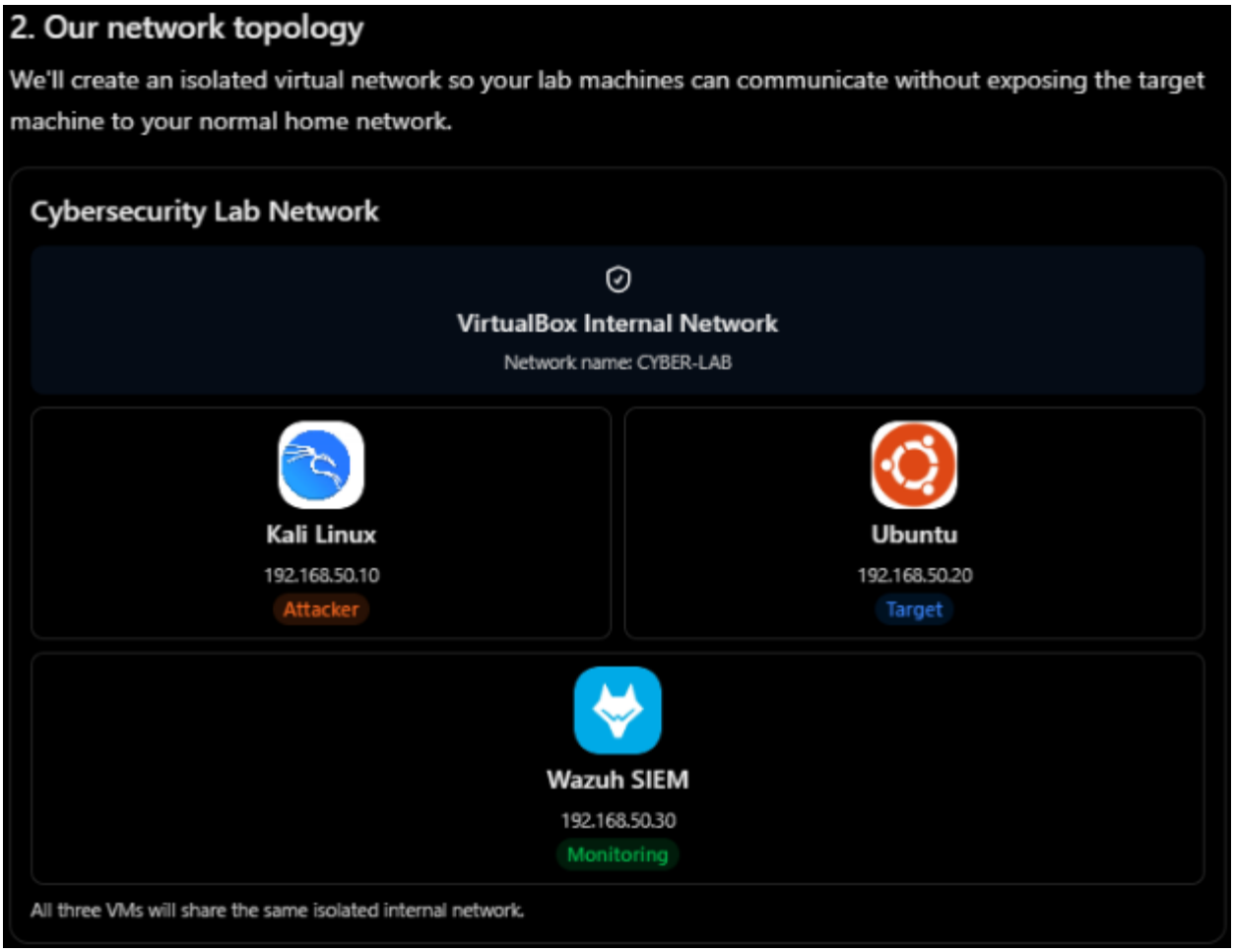
Machine	Role	IP address
Ubuntu	Main server	192.168.50.20
Kali	Security testing	192.168.50.10
Wazuh	SIEM monitoring	Initially on Ubuntu, if resources permit

We'll first verify your existing IP addresses rather than assume they haven't changed.



2. Today's implementation plan

Step	Task
1	Check network connectivity
2	Prepare Ubuntu server
3	Configure SSH securely
4	Configure UFW firewall
5	Install Wazuh
6	Connect Ubuntu to Wazuh
7	Perform security tests from Kali
8	Observe and analyze security alerts



3. Important: Configure your VirtualBox network

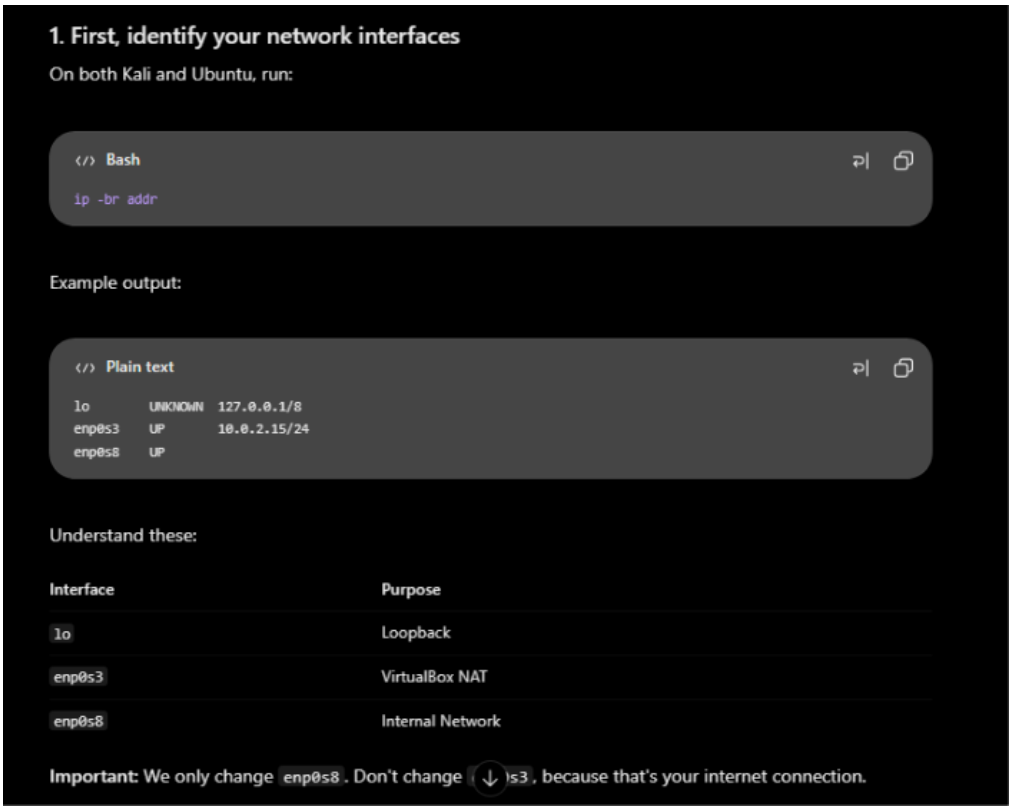
We will use two network adapters for each VM.

Adapter	Mode	Purpose
Adapter 1	NAT	Internet access for updates
Adapter 2	Internal Network	Private communication between lab VMs

Why two adapters?

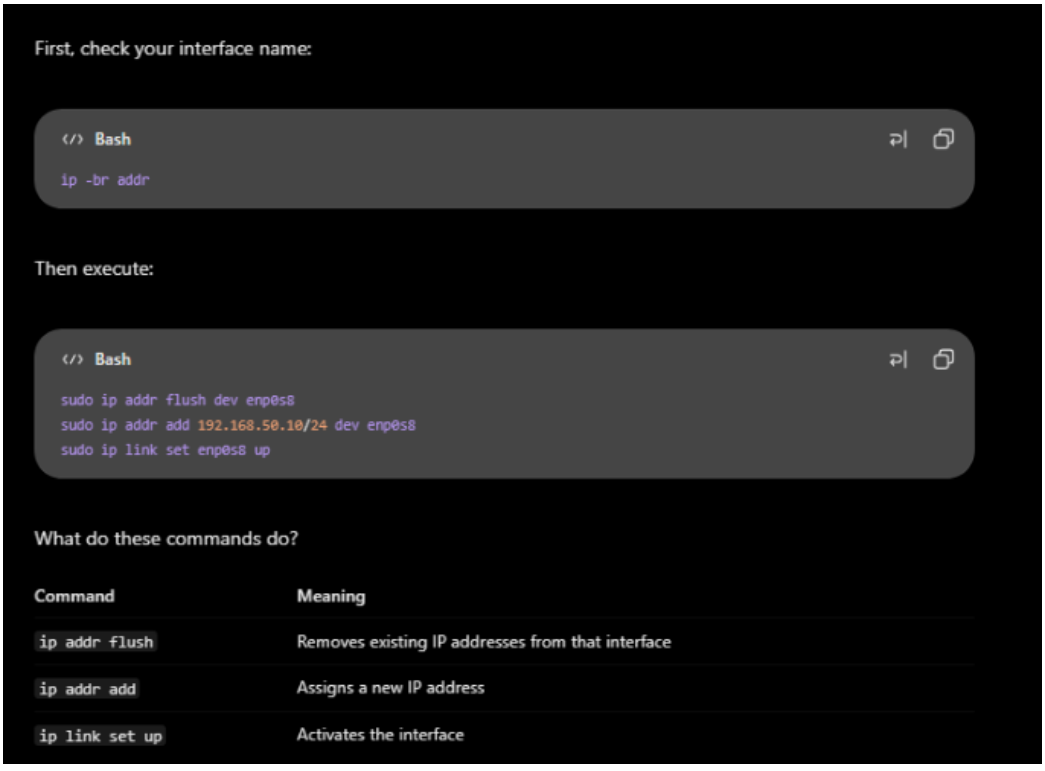
- NAT lets Kali and Ubuntu download updates.
- Internal Network allows them to communicate privately.
- We avoid Bridged networking, which could expose lab machines to your physical LAN.

4. IP CONFIG -only change the enp0s8



2. Configure Kali Linux IP address

Kali – Attacker



3. If eth1 is confirmed as your Internal Network adapter

Run:

```bash
sudo ip addr del 192.168.20.11/24 dev eth1
```

Then check:

```bash
ip -br addr
ip route
```

4. Our next practical exercise: SSH

Once the network is confirmed, we'll install SSH on Ubuntu.

On Ubuntu:

```bash
sudo apt update
sudo apt install openssh-server -y
sudo systemctl enable --now ssh
```

Check SSH:

```bash
sudo systemctl status ssh
```

On Kali:

```bash
nmap -sV -p 22 192.168.50.20
```

```
┌──(aakif㉿kali)-[~]
└─$ nmap -sV -p 22 192.168.50.20
Starting Nmap 7.98 ( https://nmap.org ) at 2026-11-01 12:39 +0530
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or sp
ecify valid servers with --dns-servers
Nmap scan report for 192.168.50.20
Host is up (0.0021s latency).

PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 9.6p1 Ubuntu 3ubuntu13.20 (Ubuntu Linux; protocol 2.0)
MAC Address: 08:00:27:24:0A:41 (Oracle VirtualBox virtual NIC)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.10 seconds

┌──(aakif㉿kali)-[~]
└─$
```

```
aakif@ubuntu: ~
Session  Actions  Edit  View  Help
└─$ ssh aakif@192.168.50.20
aakif@192.168.50.20's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 7.0.0-38-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

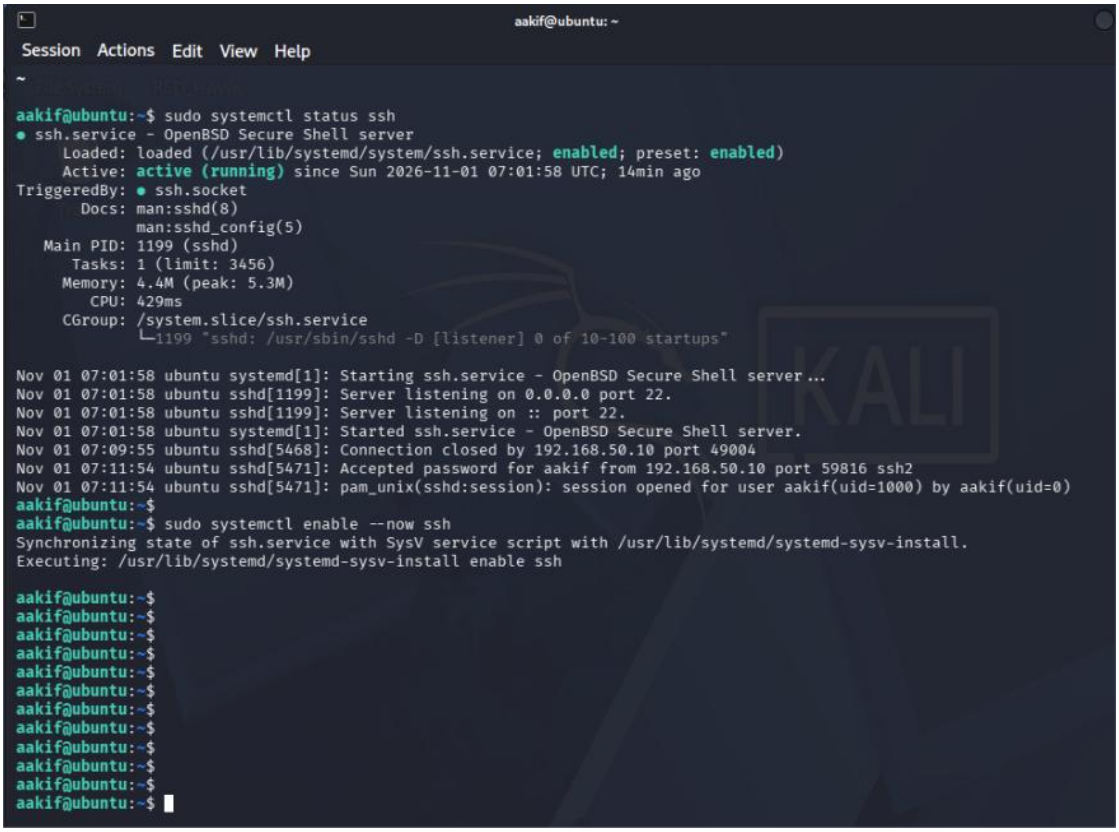
Expanded Security Maintenance for Applications is not enabled.

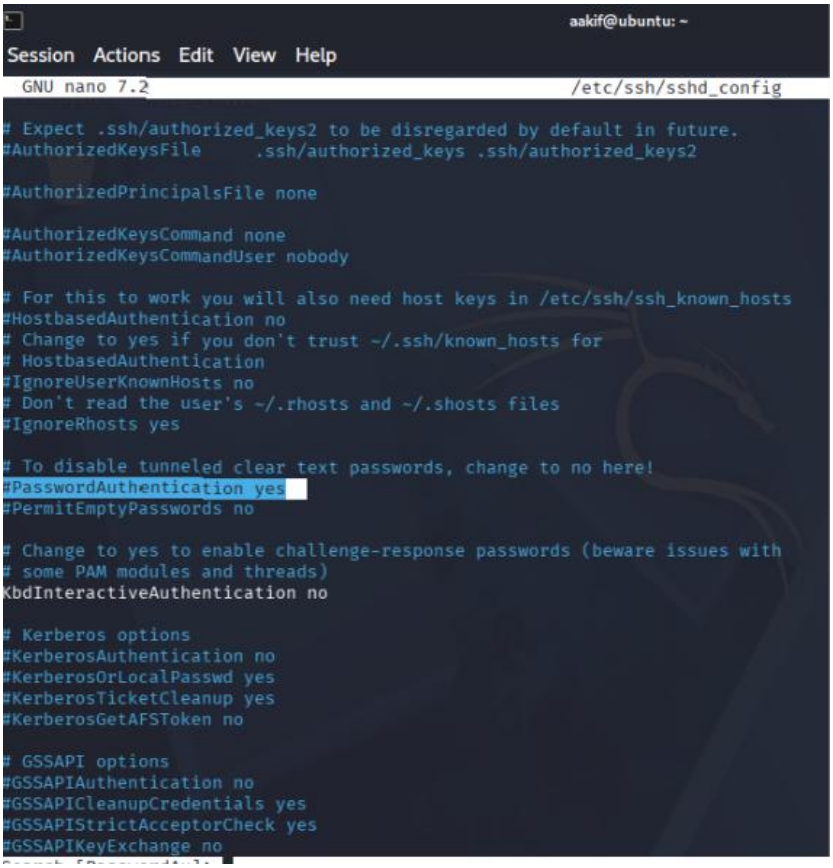
156 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

37 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Sat Oct 31 03:10:20 2026 from 192.168.50.10
aakif@ubuntu:~$ whoami
aakif
aakif@ubuntu:~$ sudo systemctl status ssh
[sudo] password for aakif:
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
     Active: active (running) since Sun 2026-11-01 07:01:58 UTC; 10min ago
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 1199 (sshd)
      Tasks: 1 (limit: 3456)
     Memory: 4.1M (peak: 5.3M)
        CPU: 429ms
     CGroup: /system.slice/ssh.service
             └─1199 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Nov 01 07:01:58 ubuntu systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Nov 01 07:01:58 ubuntu sshd[1199]: Server listening on 0.0.0.0 port 22.
Nov 01 07:01:58 ubuntu sshd[1199]: Server listening on :: port 22.
Nov 01 07:01:58 ubuntu systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Nov 01 07:09:55 ubuntu sshd[5468]: Connection closed by 192.168.50.10 port 49004
Nov 01 07:11:54 ubuntu sshd[5471]: Accepted password for aakif from 192.168.50.10 port 59816 ssh2
```


aakif@ubuntu: ~
Session Actions Edit View Help
aakif@ubuntu:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
Active: active (running) since Sun 2026-11-01 07:01:58 UTC; 14min ago
TriggeredBy: ● ssh.socket
Docs: man:sshd(8)
man:sshd_config(5)
Main PID: 1199 (sshd)
Tasks: 1 (limit: 3456)
Memory: 4.4M (peak: 5.3M)
CPU: 429ms
CGroup: /system.slice/ssh.service
└─1199 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
Nov 01 07:01:58 ubuntu systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Nov 01 07:01:58 ubuntu sshd[1199]: Server listening on 0.0.0.0 port 22.
Nov 01 07:01:58 ubuntu sshd[1199]: Server listening on :: port 22.
Nov 01 07:01:58 ubuntu systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Nov 01 07:09:55 ubuntu sshd[5468]: Connection closed by 192.168.50.10 port 49004
Nov 01 07:11:54 ubuntu sshd[5471]: Accepted password for aakif from 192.168.50.10 port 59816 ssh2
Nov 01 07:11:54 ubuntu sshd[5471]: pam_unix(sshd:session): session opened for user aakif(uid=1000) by aakif(uid=0)
aakif@ubuntu:~$
aakif@ubuntu:~$ sudo systemctl enable --now ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
aakif@ubuntu:~$
aakif@ubuntu:~$
aakif@ubuntu:~$
aakif@ubuntu:~$
aakif@ubuntu:~$
aakif@ubuntu:~$
aakif@ubuntu:~$
aakif@ubuntu:~$
aakif@ubuntu:~$
aakif@ubuntu:~$
aakif@ubuntu:~$


aakif@ubuntu: ~
Session Actions Edit View Help
GNU nano 7.2 /etc/ssh/sshd_config
Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2
#AuthorizedPrincipalsFile none
#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody
For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
Change to yes if you don't trust ~/.ssh/known_hosts for
HostbasedAuthentication
#IgnoreUserKnownHosts no
Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes
To disable tunneled clear text passwords, change to no here!
#PasswordAuthentication yes
#PermitEmptyPasswords no
Change to yes to enable challenge-response passwords (beware issues with
some PAM modules and threads)
KbdInteractiveAuthentication no
Kerberos options
#KerberosAuthentication no
#KerberosOrLocalPasswd yes
#KerberosTicketCleanup yes
#KerberosGetAFSToken no
GSSAPI options
#GSSAPIAuthentication no
#GSSAPICleanupCredentials yes
#GSSAPIStrictAcceptorCheck yes
#GSSAPIKeyExchange no

Step 3 — Install Wazuh

Once the disk is expanded and we've checked the available memory, we'll run:

```
cd ~
```

```
curl -sO https://packages.wazuh.com/4.14/wazuh-install.sh
```

```
sudo bash ./wazuh-install.sh -a
```

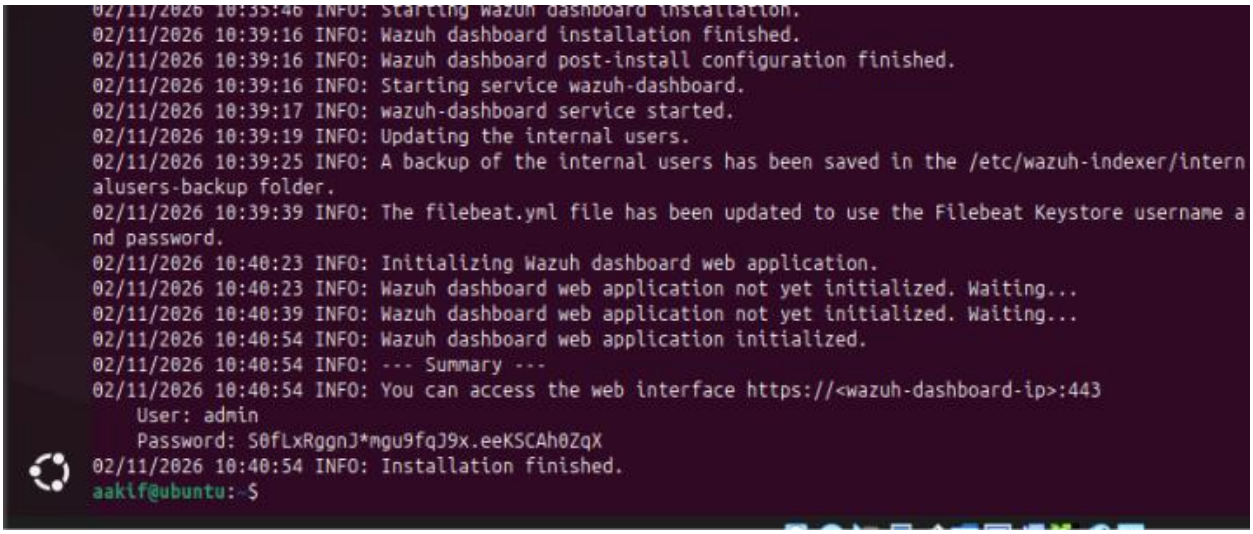
This installs all three central components:

Component	Purpose
Wazuh Indexer	Stores and searches security alerts
Wazuh Manager	Analyses events from agents
Wazuh Dashboard	Web interface for monitoring

After installation, we'll save the generated login credentials and access the dashboard at https://UBUNTU-IP.

Official reference: Wazuh Quickstart.

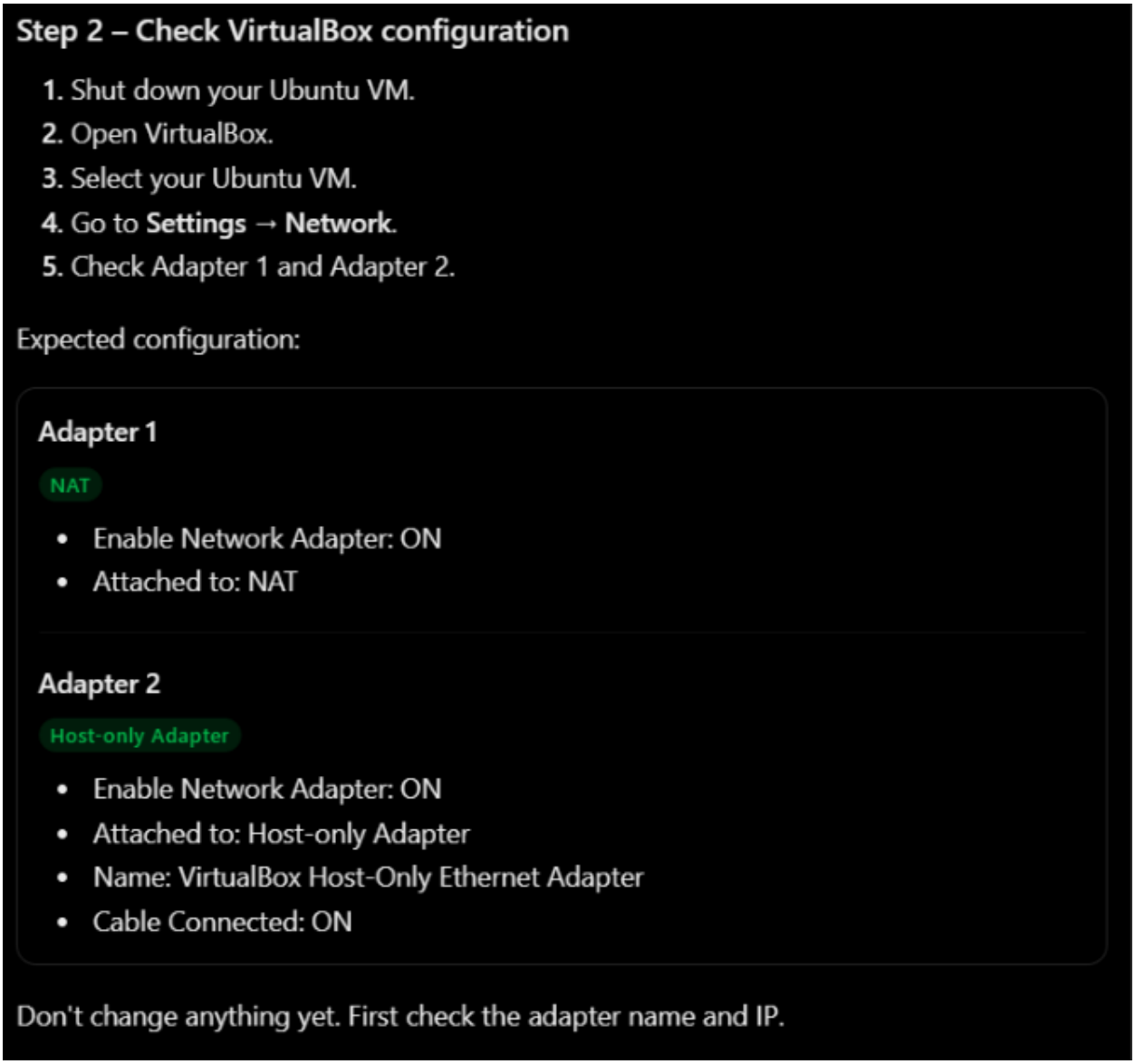
Your next task: Can you increase the Ubuntu disk to 60 GB? Send me the output of lsblk and free -h after that. We'll continue with the actual installation together.



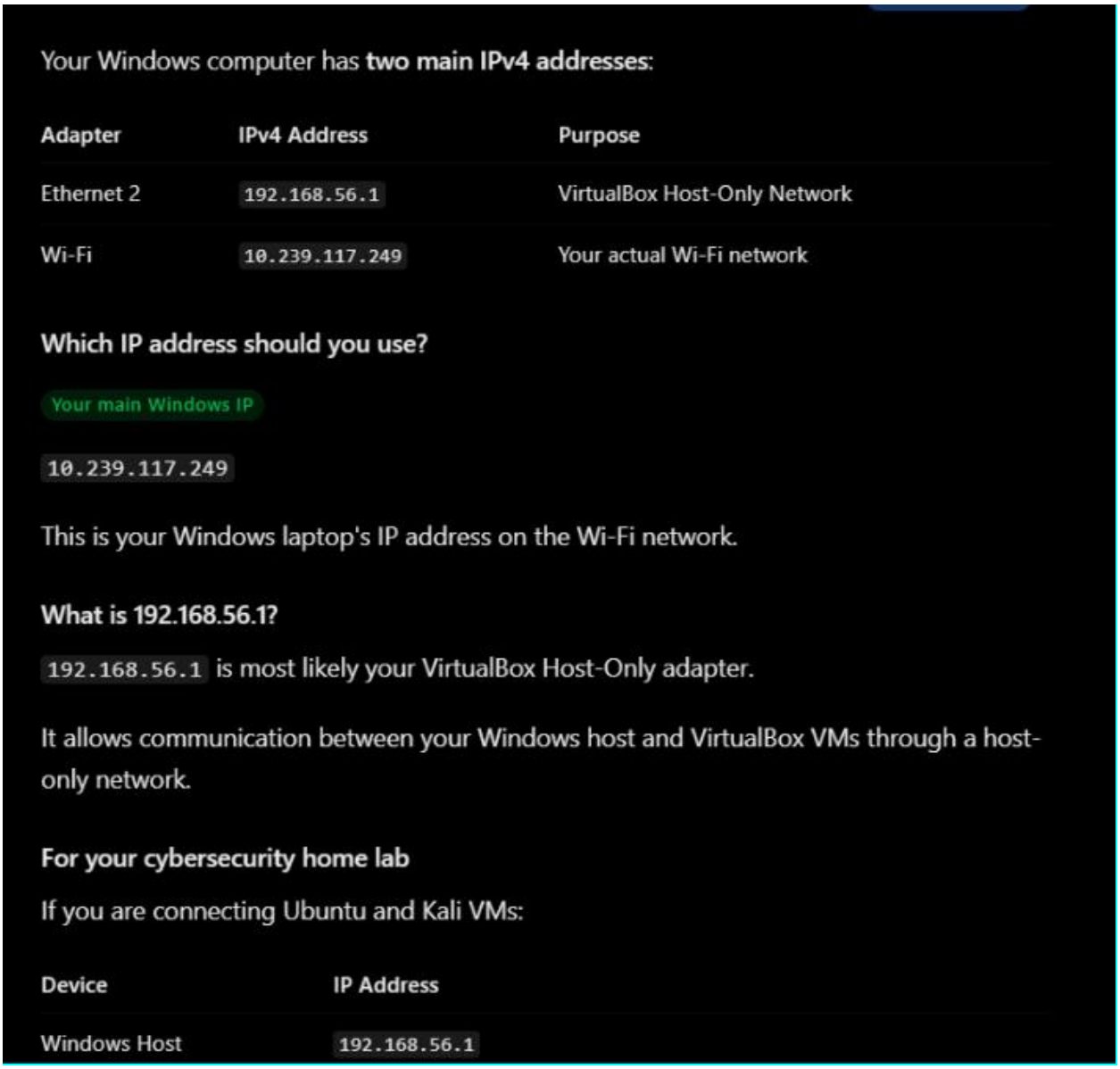
User: admin

Password: S0fLxRggnJ*mgu9fqJ9x.eeKSCAh0ZqX

Checking the adapter



Windows ip – 192.168.56.1



Your Windows computer has **two main IPv4 addresses**:

Adapter	IPv4 Address	Purpose
Ethernet 2	`192.168.56.1`	VirtualBox Host-Only Network
Wi-Fi	`10.239.117.249`	Your actual Wi-Fi network

Which IP address should you use?

Your main Windows IP

`10.239.117.249`

This is your Windows laptop's IP address on the Wi-Fi network.

What is 192.168.56.1?

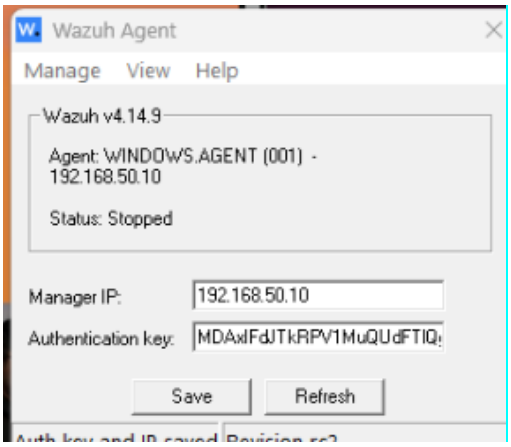
`192.168.56.1` is most likely your VirtualBox Host-Only adapter.

It allows communication between your Windows host and VirtualBox VMs through a host-only network.

For your cybersecurity home lab

If you are connecting Ubuntu and Kali VMs:

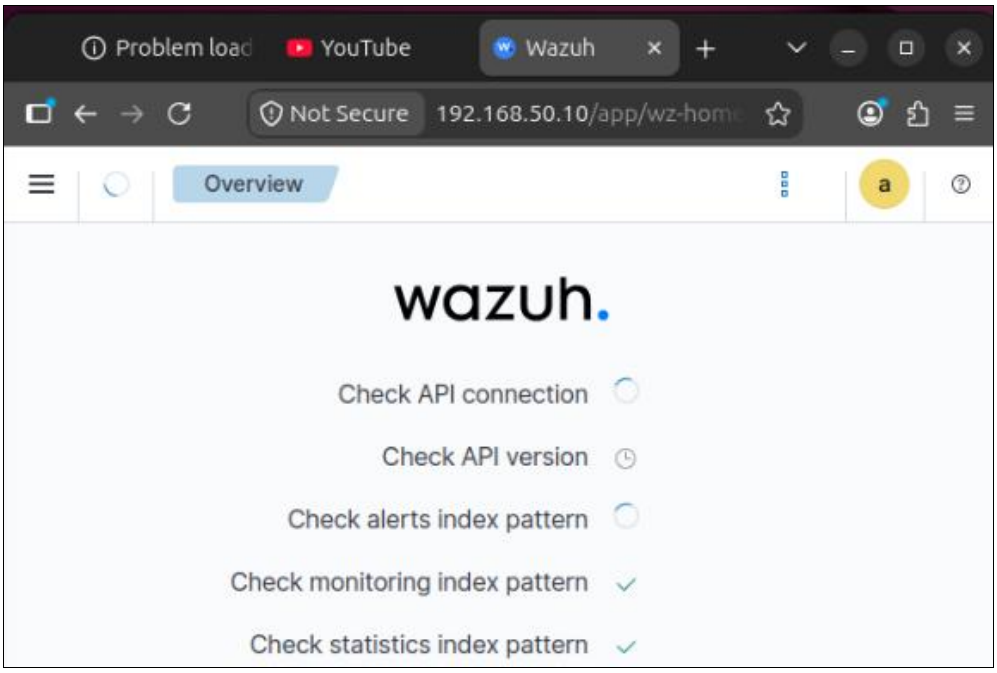
Device	IP Address
Windows Host	`192.168.56.1`



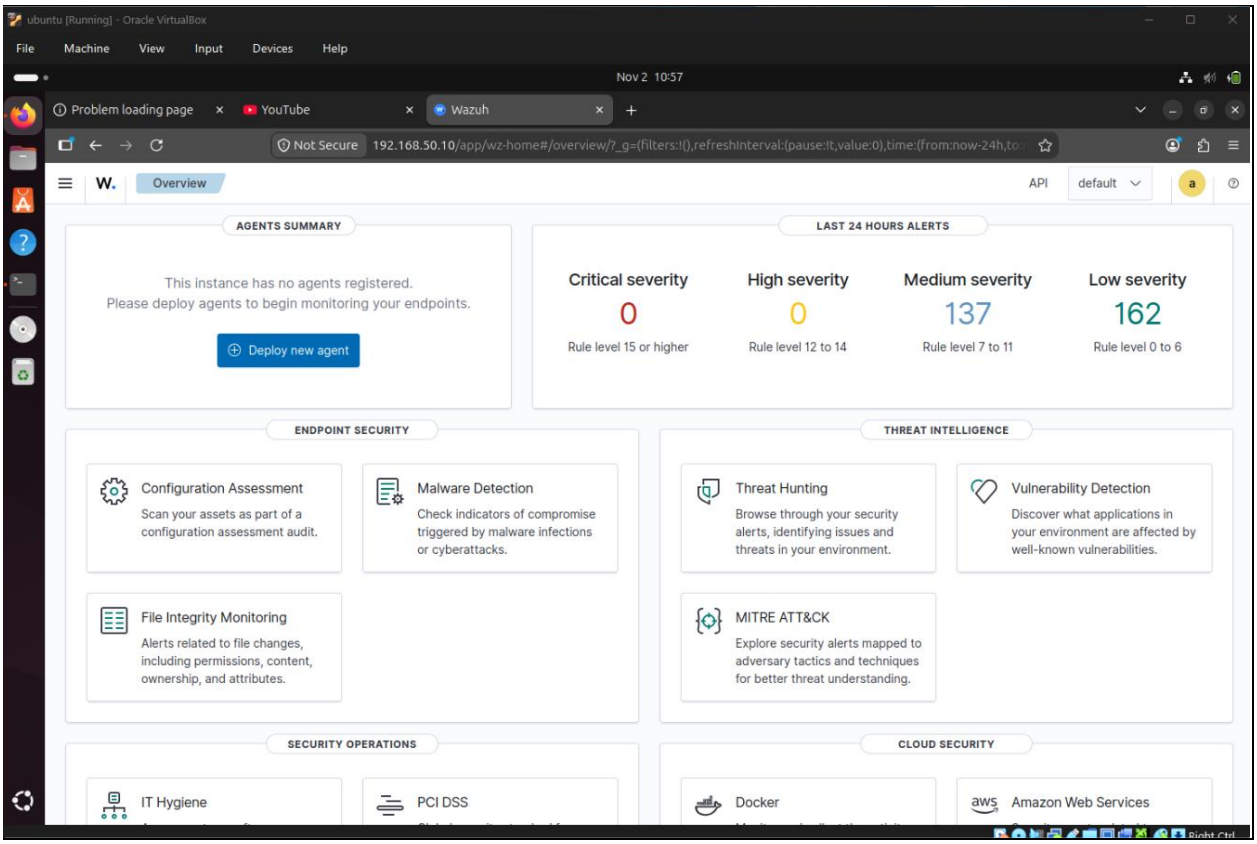
SO GO TO MANGE AND SELECT THE RESTART TO RUN IT

All things fine now open wazuh so in web browser type = 192.168.50.10 so this ip go to wazuh main page





Wazuh home page now seeing



6. Installing the Wazuh Agent (Windows Host)

1. Download the latest Wazuh agent MSI installer from the official documentation:

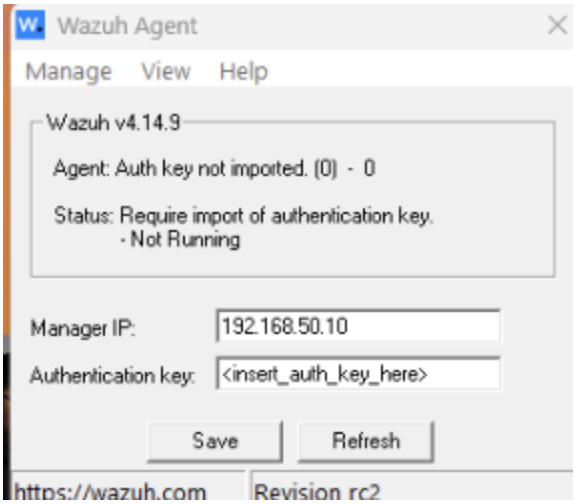
Wazuh Agent for Windows – this is the side

Download windows installer to start the installation

2. Install the MSI pac

kage on your Windows system using the default settings.

Install it then ip



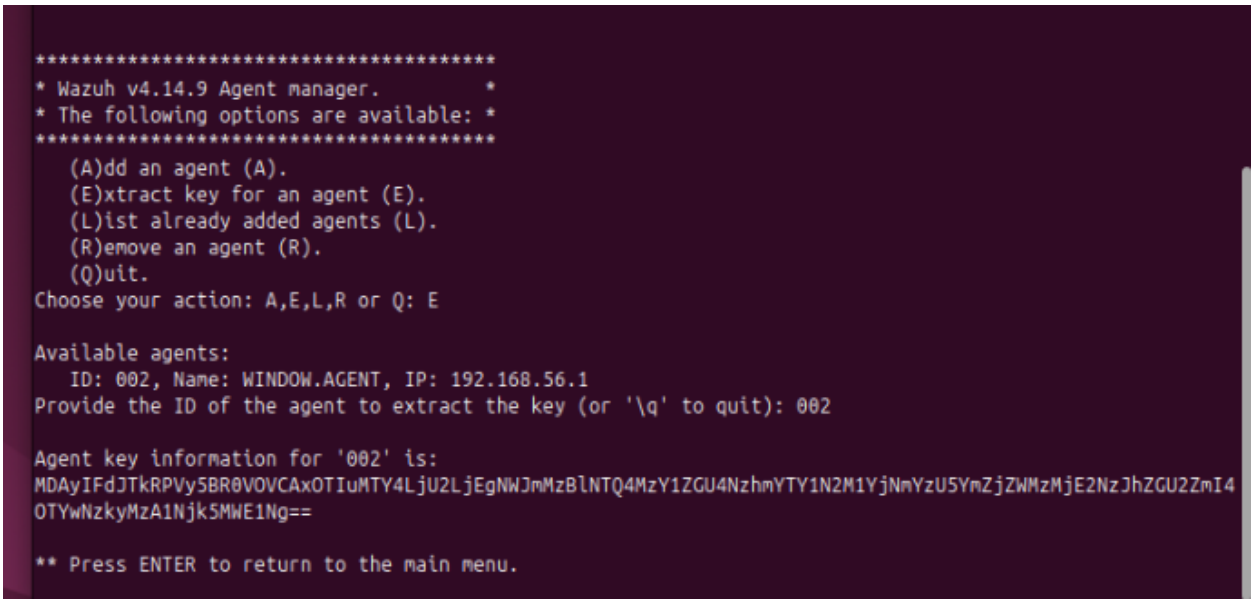
Find the auth key

7. Registeringthe Agent with the Manager

7.1 Generate Agent Key on Ubuntu Manager Run the agent management utility:

```
sudo /var/ossec/bin/manage_agents
```

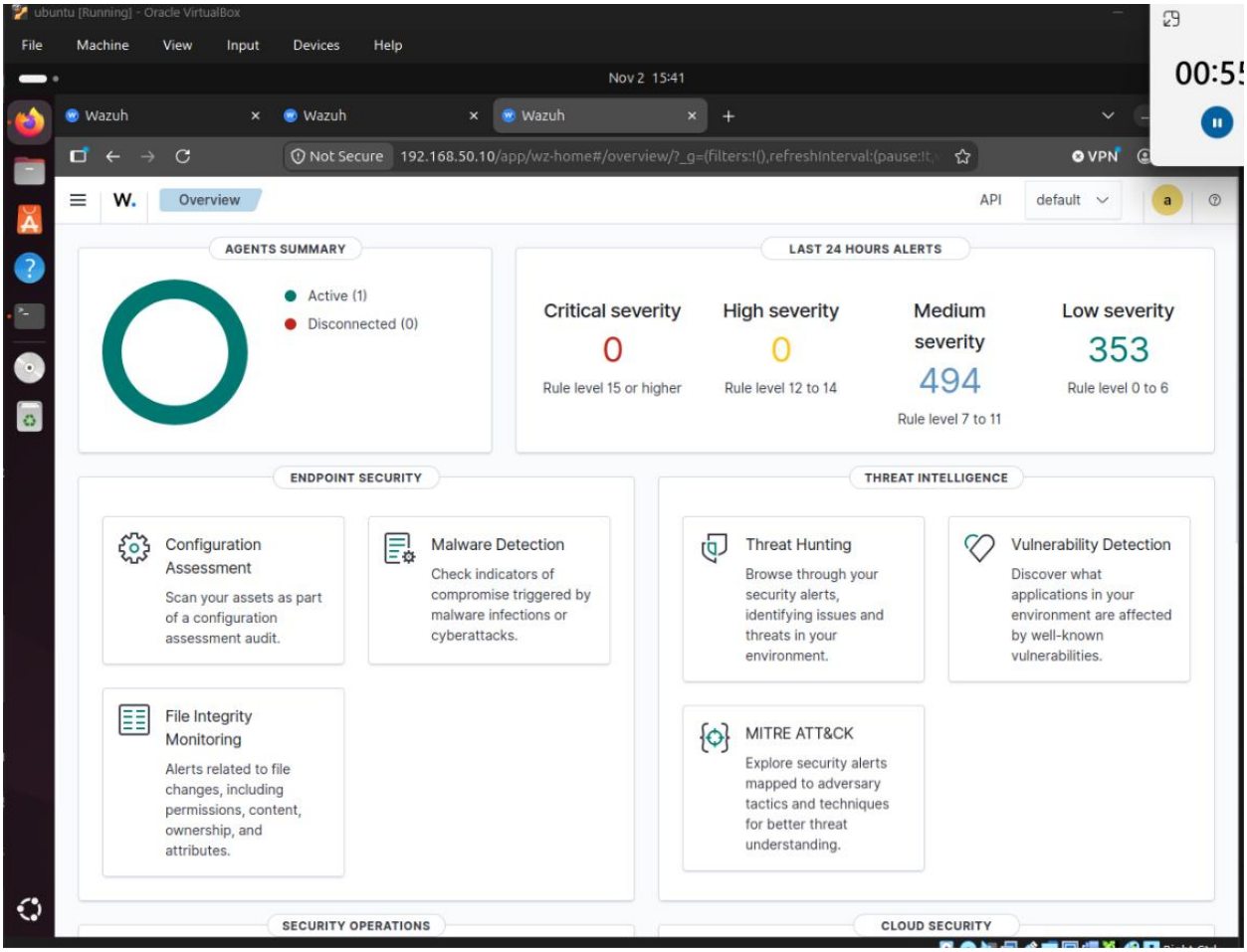
- Select A to add an agent.
- Assign a name (e.g., WindowsHost).
- Leave IP address blank unless static assignment is needed.
- After creation, select E to extract the key.
- Copy the key output



Name WINDOWS.AGENT
IP AGENT :192.168.50.10
Id : 001
AGAIN SELECT : E
THEN THE ID IS :001
THEN YOU HAVE THE KEY –

MDAyIFdJTkRPVy5BR0VOVCAxOTIuMTY4LjU2LjEgNWJmMzBlNTQ4MzY1ZGU4NzhmYTY1N2M1YjNmYzU5YmZjZWMzMjE2NzJhZGU2ZmI4OTYwNzkyMzA1Njk5MWE1Ng==

Dashboard



Here ip ubuntu = 192.168.50.10

Ip for windows = 192.168.50.1

Ping successfully

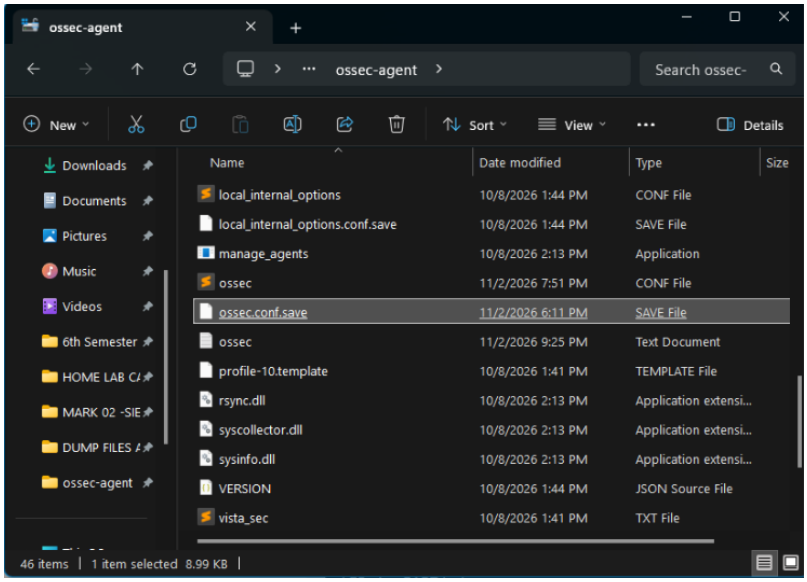
8. File Integrity Monitoringon Windows

Wazuh supports real-time monitoring of file and folder changes using Syscheck.

8.1 Edit Agent Configuration

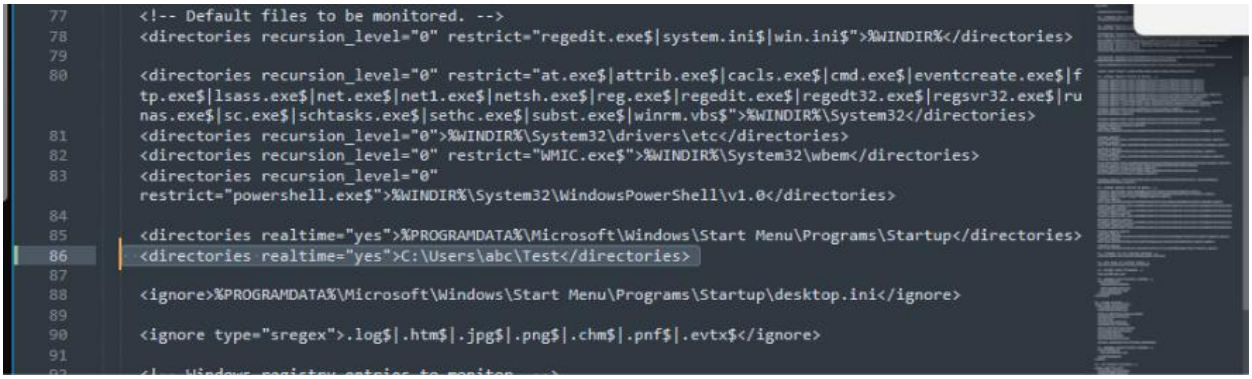
- Open the following configuration file:

  C:\Program Files (x86)\ossec-agent



- Add the following entry inside the Directory block: this would be the path the file activities can accessible to see

<directories realtime="yes">C:\Users\abc\Test</directories>



This isn't access to change the code here ask permission so

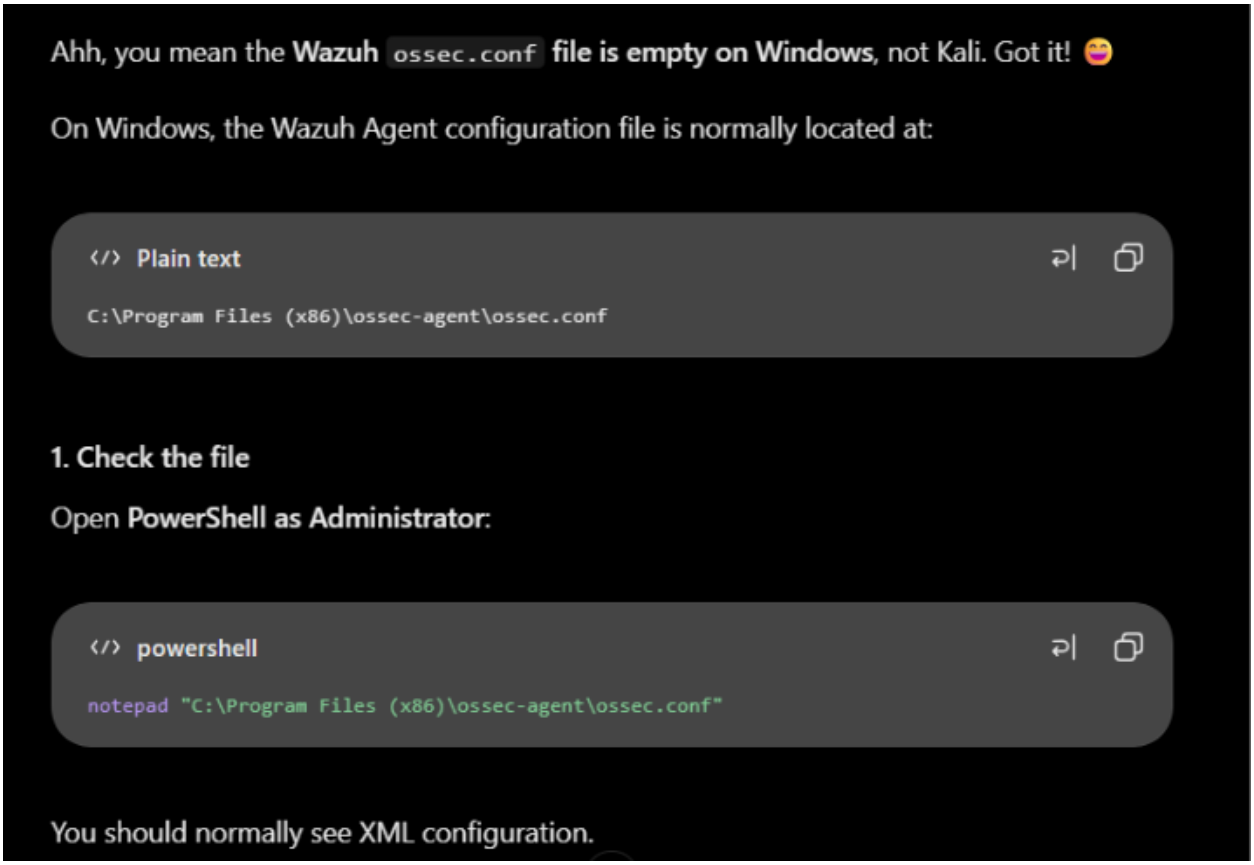
On Windows, the Wazuh Agent configuration file is normally located at:
C:\Program Files (x86)\ossec-agent\ossec.conf

1. Check the file

Open **PowerShell** as **Administrator:**

notepad "C:\Program Files (x86)\ossec-agent\ossec.conf"

You should normally see XML configuration.

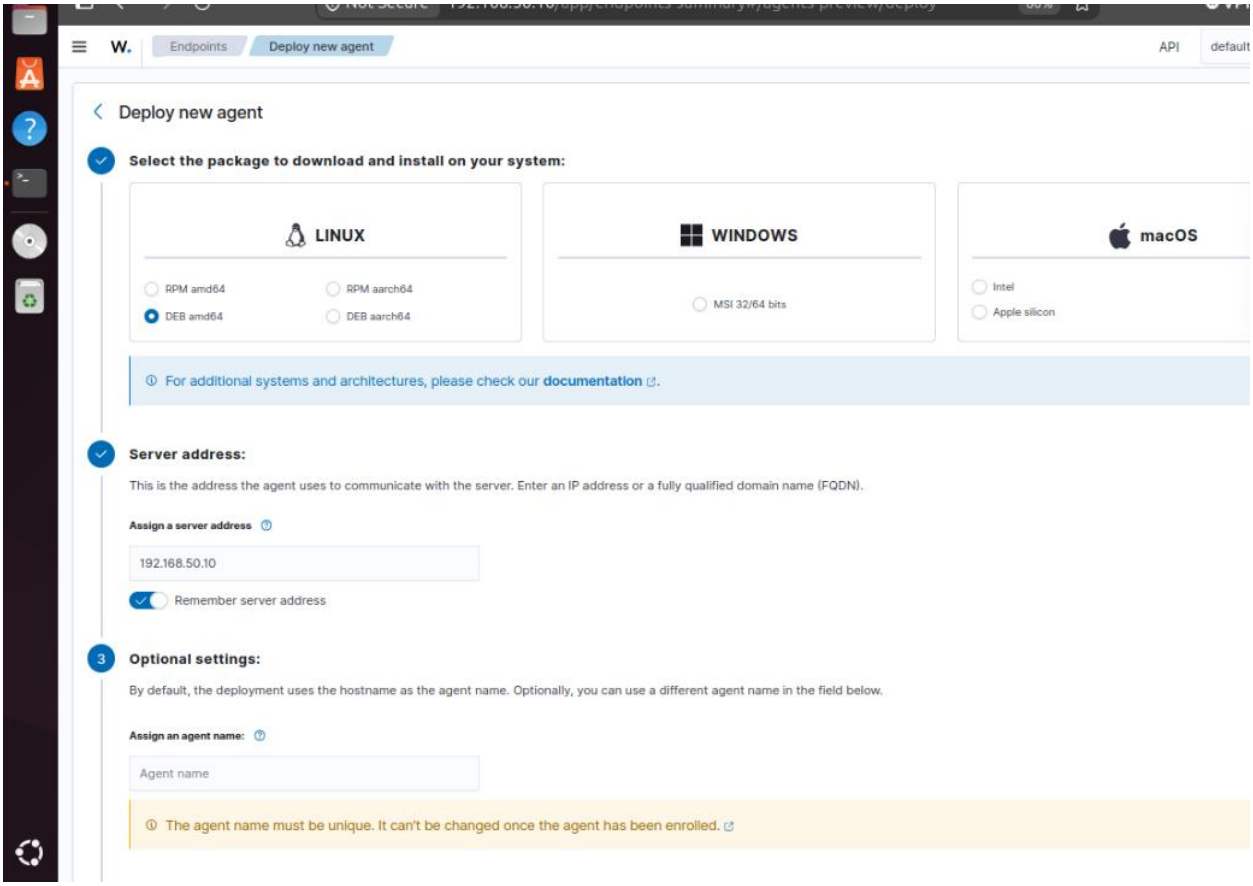


Path –
<directories realtime="yes">C:\New Downloaded Programing Files\Wazuh dir</directories>

Ok then restart agent again (WAZUH AGENT windows)

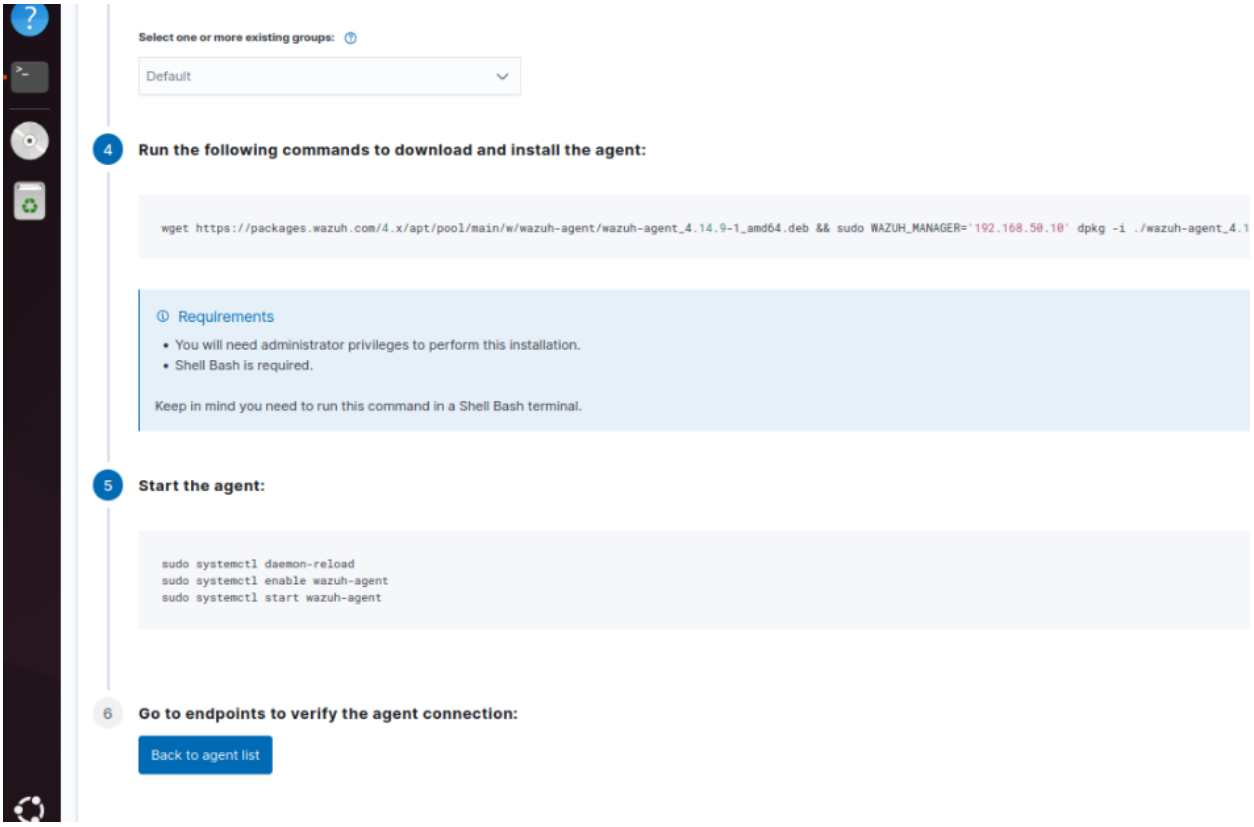
Founded easy way to add to wazuh

https://youtu.be/bltbJ2TUQWU?si=PBB8pC7j4ICY-EOz



Next copy the text and run as a administrator

Sudo bash - then its privileged mode so can add now



2.1 – Prepare Ubuntu SSH monitoring

Open your Ubuntu terminal.

Check SSH service:

```
sudo systemctl status ssh
```

If it isn't installed, run:

```
sudo apt update
sudo apt install openssh-server -y
sudo systemctl enable --now ssh
```

Check Ubuntu's lab IP:

ip -br addr

We need the IP address on your isolated 192.168.50.0/24 network.

2.1 – Prepare Ubuntu SSH monitoring

Open your **Ubuntu terminal**.

Check SSH service:

```bash
sudo systemctl status ssh
```

If it isn't installed, run:

```bash
sudo apt update
sudo apt install openssh-server -y
sudo systemctl enable --now ssh
```

Check Ubuntu's lab IP:

```bash
ip -br addr
```

We need the IP address on your isolated `192.168.50.0/24` network.

SSH Brute-Force Attack Detection

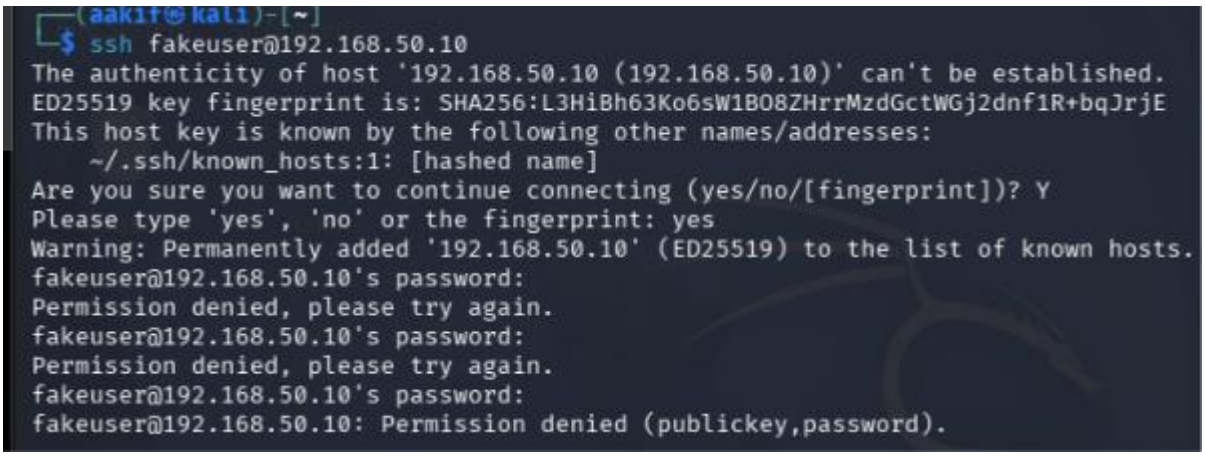
PRACTICAL SOC EXPERIMENT 01

2.3 – Simulate failed logins from Kali

Open your Kali terminal.

- Replace 192.168.50.10 below with your Ubuntu IP if different.
  ssh fakeuser@192.168.50.10

- Enter an incorrect password when prompted
- Repeat this a few times, for example 5 attempts.
- If the SSH client disconnects, reconnect and continue.
- Important: Use only your own Ubuntu lab machine and don't use external attack tools or external targets.



2.4 – Investigate the events in Wazuh

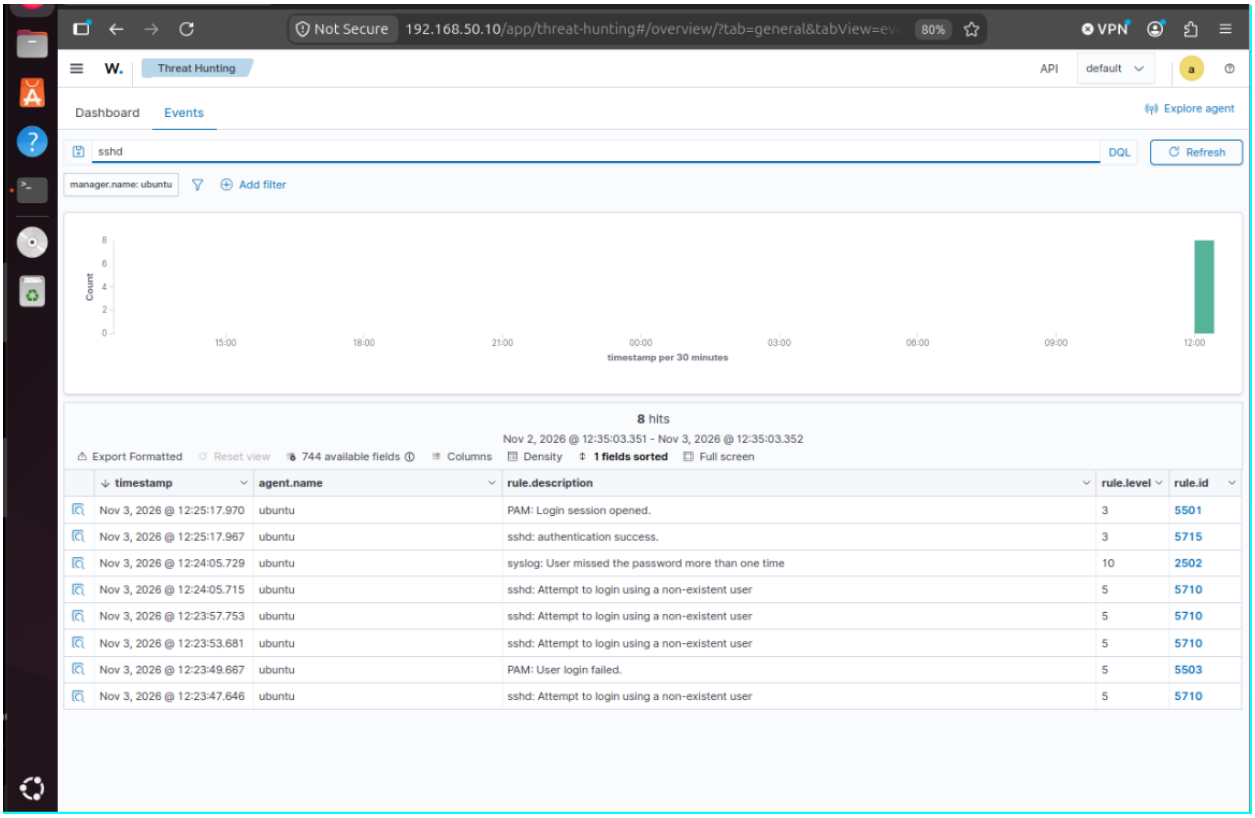
Open your Wazuh Dashboard on Ubuntu.

Navigate to:

Threat Hunting → Events

Search for:

Sshd



Also try:

authentication failed

Look for events containing:

- SSH authentication failure.
- Invalid user.
- Failed password.
- Source IP address of Kali.

Note: A failed login event may appear before Wazuh generates a higher-level brute-force alert. We will configure a detection rule if needed.

Configure SSH Brute-Force Detection

SOC EXPERIMENT 02

Hydra SSH Attack Detection

Practical Experiment 04 – Hydra SSH Attack Detection

Authorized home lab only

Step 1 – Prepare Ubuntu (target machine)

Open your Ubuntu terminal.

1.1 Check SSH service

```
sudo systemctl status ssh
```

It should show:
Active: active (running)

1.2 Create a dedicated test account

```
sudo adduser hydratest
```

Set a temporary lab password when prompted. Don't use your real password.

1.3 Confirm Ubuntu IP

```
ip -br addr
```

Find the IP on your isolated lab network, likely `192.168.5010`.

1.4 Monitor SSH logs

Open a second Ubuntu terminal:

```
sudo journalctl -u ssh -f
```

Keep this running during the experiment.

Step 2 – Install Hydra on Kali

Open your Kali terminal, not Ubuntu.
sudo apt update
sudo apt install hydra -y

Verify installation:
hydra -h

You should see Hydra usage information.

[image](https://www.google.com/s2/favicons?domain=https://www.kali.org\&sz=32)

Kali Linux Tools+1

Step 3 – Create a small password list

On Kali
nano lab-passwords.txt

Enter these eight deliberately incorrect test passwords:

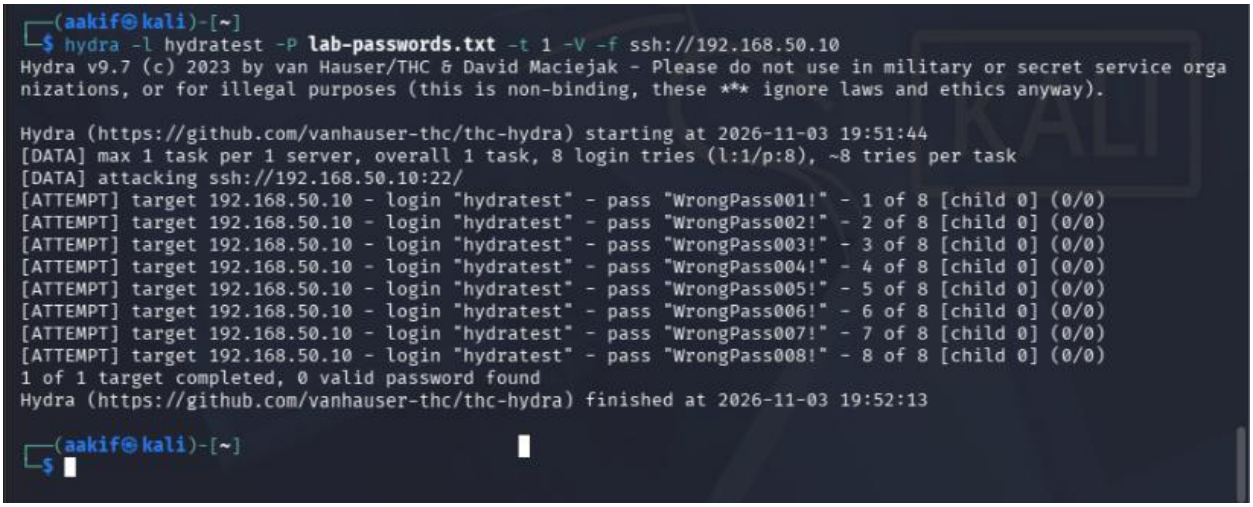
WrongPass001!
WrongPass002!
WrongPass003!
WrongPass004!
WrongPass005!
WrongPass006!
WrongPass007!
WrongPass008!

Save with `Ctrl + O`, Enter, then `Ctrl + X`.

Check the list:
wc -l lab-passwords.txt

Expected:
8 lab-passwords.txt

The actual password of `hydratest` must not be in this list.



Step 4 – Confirm your network

On Kali:
ip -br addr

On Ubuntu:
ip -br addr

Step 5 – Run the controlled Hydra test

Only run against your own Ubuntu VM

On Kali, use:
hydra -l hydratest -P lab-passwords.txt -t 1 -V -f ssh://192.168.50.20

Replace the IP if your Ubuntu address is different.

What the options mean:

Option	Meaning
`-l`	Single username
`-P`	Password list
`-t 1`	One connection at a time
`-V`	Show each attempt
`-f`	Stop if valid credentials are found
`ssh://`	SSH service

This is deliberately limited to eight attempts against your dedicated lab account.

Step 6 – Observe the attack from Ubuntu

While Hydra runs, open your Ubuntu terminal:
sudo journalctl -u ssh -f

You should see failed authentication messages.

Step 7 – Investigate in Wazuh Dashboard

Open your Ubuntu browser:

`https://localhost`

Navigate to Threat Hunting → Events.

Search: rule.groups:sshd